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(1) Research Program/Project/Study Title: IMPROVEMENT OF SWINE WELFARE: ADDRESSING GAPS IN GOOD ANIMAL HUSBANDRY PRACTICES (GAHP) AMONG SMALLHOLDER FARMERS IN PONTEVEDRA, CAPIZ, PHILIPPINES	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☒ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center	
(4) Site of implementation			
Municipality/City:		PONTEVEDRA	
Province:		CAPIZ	
Region:		VI	
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization		(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☒ Collaborative Research (external agency)	
(7) Sector/Commodity AGRICULTURE / LIVESTOCK (SWINE)		(8) Discipline/Field of Specialization ANIMAL SCIENCE	

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(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

It is critical to assess the extent to which our government authorities were able to provide information to the general public and effectively implement GAHP standards, particularly among local swine farmers. This study aimed to investigate the application of Good Animal Husbandry Practices (GAHP) among smallholder swine farmers in Pontevedra, Capiz, identifying gaps and proposing evidence-based interventions to improve attitudes and decision-making towards animal welfare. A mixed-methods approach was employed, involving 30 smallholder swine farmers for a descriptive cross-sectional survey, nine (9) for qualitative interviews or focus group discussions, and five local government officials and stakeholders for key informant interviews. Data analysis included frequency counts, percentages, and weighted means.

Findings revealed that while stakeholders recognized GAHP's benefits for animal welfare, production, disease prevention, and food safety, significant challenges hinder its full implementation. Local government units (LGUs) face financial constraints, limited technical staff, and competing priorities, which restrict effective GAHP promotion. Farmers also face challenges such as gaps in technical knowledge and skills, reliance on traditional methods, and financial constraints affecting essential areas like feed management, pest control, health care, and sanitation.

To facilitate GAHP adoption, the study suggests strengthening LGUs' capacity through increased leadership commitment, collaboration among government agencies, including private sector is crucial for providing financial, technical, and human resources support. Emphasizing GAHP principles in local codes and regulations, enhancing stakeholder engagement, and improving farmer awareness through regular orientations and incentives are recommended. Organizing farmers into cooperatives can further promote GAHP adoption. Mechanisms for regular monitoring and evaluation of GAHP implementation is also essential for assessing policy effectiveness and identifying areas for improvement.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Good Animal Husbandry Practices (GAHP); Backyard Swine Farming; Animal Welfare in Swine; Smallholder Farmers; Pontevedra, Capiz; Mixed-Methods Research

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

The welfare of farm animals is considered integral to animal product quality (Liang et al., 2022). However, concerns about the welfare of these animals can be compromised due to factors such as a farmer's capacity, drive, and goals. Farmers' attitudes toward animal welfare and their decisions regarding it are influenced by their values, beliefs, and experiences (Vigors et al., 2021). The Animal Welfare Act of 1998, with amendments by RA 10631, defines animal welfare as the physical and psychological well-being of animals, which includes avoiding abuse, maltreatment, cruelty, and exploitation of animals by maintaining appropriate standards of accommodation, feeding, and general care, preventing and treating diseases, ensuring freedom from fear, distress, harassment, and unnecessary discomfort and pain, and allowing animals to express normal behavior. In the realm of animal production in our country, particularly in the swine industry, a significant portion is managed by backyard Filipino farmers. According to data from the Philippine Statistics Authority (PSA) during the 3rd quarter of 2023, smallholder farmers account for 67.5% of the swine population, while commercial and semi-commercial farms represent 29.2% and 3.3%, respectively. This highlights the crucial role of smallholders in the livestock industry and their contribution to the rural economy's balanced growth, benefiting less fortunate segments of society.

However, despite their substantial contribution, government policies promoting animal welfare primarily target commercial and semi-commercial farms. This oversight raises concerns as smallholders, being the largest contributors, may unintentionally engage in practices that could harm animals. While efforts have been made to include them in advocacy programs, more comprehensive measures are needed to ensure their compliance with animal welfare standards. As evidenced by the 2020 Voiceless Animal Cruelty Index, which ranked the Philippines 19th globally, there is a pressing need for improved animal protection legislation. The country received a performance rating of "E," indicating a necessity for reassessment, realignment, and revision of existing regulations to enhance animal welfare. This entails a concerted effort from the government to update legislation, provide education and enforcement support, and promote welfare of animals.

To promote good animal husbandry practices facilitating of animal welfare and food safety, the Bureau of Agriculture and Fisheries Product Standards (BAFPS) developed the CODE of Good Animal Husbandry Practices (GAHP), which provides guidelines for good farming practices and minimal standards for rearing animals commercially or in one's backyard for food production. Despite the existing rules and regulations, it is essential to provide relevant services at the local level to facilitate food quality while considering animal welfare. Regarding the necessity of putting GAHP into practice, the majority of hog raisers in Capiz, as well as other provinces in Western Visayas, are backyard farmers. In the event of a pork supply shortage due to the African Swine Fever (ASF) outbreak, the cost of pork, the primary protein source for Filipinos, rises immediately (Espino et al., 2022). Because there is currently no cure for ASF, the Department of Agriculture - Bureau of Animal Industry recommends strictly following biosecurity measures, which are part of Good Animal Husbandry Practices (GAHP), to prevent the illness from spreading. Furthermore, it is critical to assess the extent to which our government authorities were able to provide information to the general public and effectively implement GAHP standards, particularly among local swine farmers. Hence, the main goal of this study is to investigate the application of Good Animal Husbandry Practices (GAHP) among local swine raisers in selected barangays in Pontevedra, Capiz, and identify gaps for evidence-based interventions to help improve attitudes and decision-making with consideration to animal welfare.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

Methodology

This study adopted a mixed-methods approach to investigate GAHP implementation among smallholder swine farmers in Pontevedra, Capiz. By combining qualitative and quantitative methods, a comprehensive understanding of the research objectives was achieved. This study was conducted from September 2023 to March 2024. The final survey was conducted at selected barangays of Pontevedra, Capiz. The focus group discussion was conducted at Capiz State University, and the key informant interview was conducted in selected areas in the Province of Capiz. This research consisted of randomly 30 smallholder swine farmers as respondents of the study's descriptive cross-sectional survey, nine selected respondents from smallholder swine farmers for Qualitative Interviews or Focus Group Discussions (FGD), and five representatives from local government officials and stakeholders for Key Informant interviews (KII).

A. Descriptive Cross-sectional Survey

Objective: Describe the current husbandry practices of swine backyard farmers.

Methodology: Developed and administered a structured survey questionnaire to a representative.

B. Qualitative Interviews or Focus Group Discussions

Objective: Gain qualitative insights into the knowledge, experiences, and capacities of swine farmers regarding GAHP.

Methodology: Conducted comprehensive interviews or focus group discussions with selected swine farming communities.

C. Key Informant Interviews

Objective: Assess the capacity and effectiveness of local government units and stakeholders in promoting GAHP.

The statistical tools used in analyzing the final survey data for quantitative statistics were frequency count, percentages and the weighted mean.

Frequency count. This was used to determine the distribution of values and to identify patterns and trends from respondents' responses.

Percentages. This was used to express in terms of a ratio that represents a fraction of 100 to compare and contrast one quantity against another.

Weighted mean. This was used to determine participants' averages or common responses during the initial survey.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Gaps in Implementing Good Animal Husbandry Practices for Swine in Pontevedra, Capiz
Suitability and Constraints:

Stakeholders view Good Animal Husbandry Practices (GAHP) positively, recognizing their benefits for animal welfare, swine production, disease prevention, and food safety. However, challenges include inadequate infrastructure, technical operations, and financial constraints among smallholder farmers. Limited resources hinder record-keeping, veterinary care, and biosecurity implementation. Comparing new and traditional methods is essential for GAHP adoption. Despite efforts by local government units (LGUs), many swine farmers remain uncooperative, affecting trust and confidence.

Most LGUs, including Pontevedra, lack active GAHP promotion due to financial constraints, multitasking staff, and reliance on national funding. Strengthening LGU leadership and collaboration with provincial and national agencies is crucial. Hiring livestock technicians within LGUs can provide necessary technical support. Effective local market regulation is needed for successful GAHP implementation.

Many farmers rely on traditional methods and lack technical knowledge required by GAHP. Education level influences their decision-making, and they often imitate other farmers' techniques. They possess basic management skills but struggle with record-keeping and biosecurity. Increasing technical knowledge, financial capacity, and confidence through targeted programs is essential. LGUs play a vital role in providing training, incentives, and support.

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Key gaps include technical knowledge, financial support, and understanding of disease transmission and biosecurity. Capacity-building programs, financial aid, and practical, affordable pest management options are needed. Improved veterinary services and technical training can enhance farm management. Sustainable practices and government-backed programs can support long-term GAHP adoption.

Strong leadership commitment and local ordinances supporting GAHP principles are essential. LGUs should create training programs, hire qualified staff, and allocate adequate funds. Establishing livestock technicians and focal persons in each barangay can improve technical assistance. Forming cooperatives can enhance GAHP implementation. Proposed projects like a "Livestock Center" can regulate market activities and support backyard raisers. Developing adaptable local protocols aligned with GAHP and creating demonstration farms can promote experiential learning and effective GAHP adoption.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

I extend my sincere thanks to Animal Empathy Philippines (AEP) for their generous funding, which made this research possible. I also express my deep gratitude to the key officials of Capiz State University for their invaluable support in making this initiative a success.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indention with 12-point space in between.)

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